

Module 1-CSS3

Theme: Styling the "Local Community Event Portal"

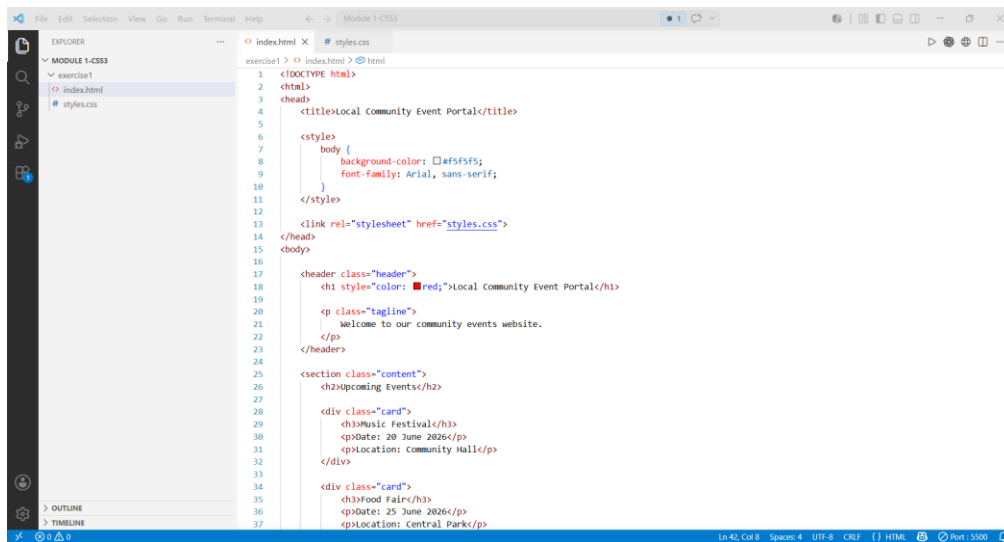
1. Why CSS? Inline vs. Internal vs. External

Scenario: The designer wants you to experiment with different ways to apply styles.

Objective: Understand various CSS inclusion methods and their impact.

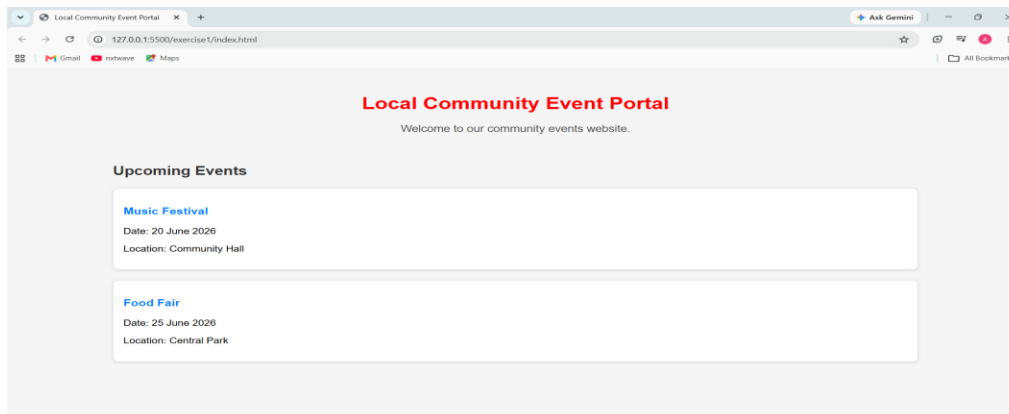
Task:

- Apply an inline style to make one heading red.
- Use an embedded to define body background.
- Link an external stylesheet styles.css and move all reusable styles there.
- Add comments in your CSS to label each section (`/* Header styles */`)



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Local Community Event Portal</title>
5
6 <style>
7   body {
8     background-color: #f5f5f5;
9     font-family: Arial, sans-serif;
10  }
11 </style>
12
13 <link rel="stylesheet" href="styles.css">
14 </head>
15 <body>
16
17 <header class="header">
18   <h1 style="color: red;">Local Community Event Portal</h1>
19
20   <p class="tagline">
21     Welcome to our community events website.
22   </p>
23 </header>
24
25 <section class="content">
26   <h2>Upcoming Events</h2>
27
28   <div class="card">
29     <h3>Music Festival</h3>
30     <p>Date: 20 June 2026</p>
31     <p>Location: Community Hall</p>
32   </div>
33
34   <div class="card">
35     <h3>Food Fair</h3>
36     <p>Date: 25 June 2026</p>
37     <p>Location: Central Park</p>
38   </div>
39 </section>
40 </body>
41 </html>
```

Output:



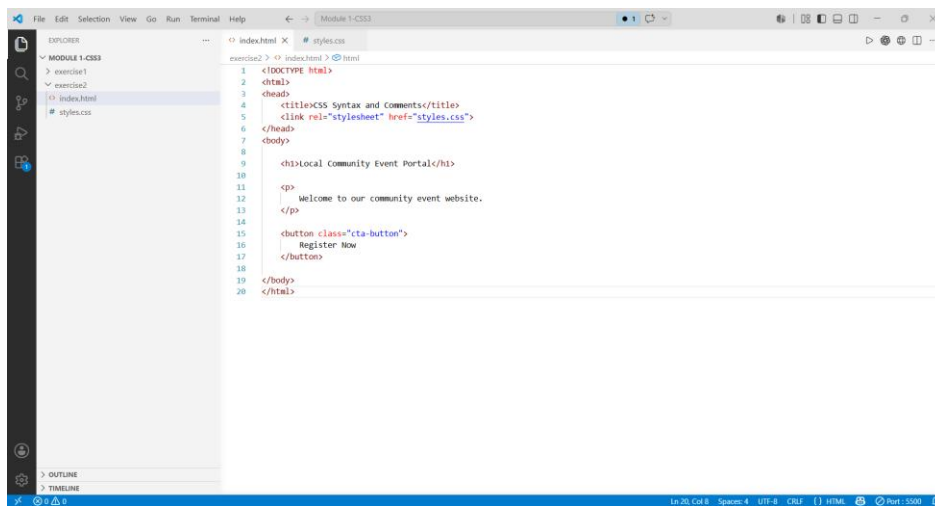
2. CSS Syntax and Comments

Scenario: You've joined a team and need to understand and maintain a large stylesheet.

Objective: Write clean, readable CSS with proper structure and comments.

Task:

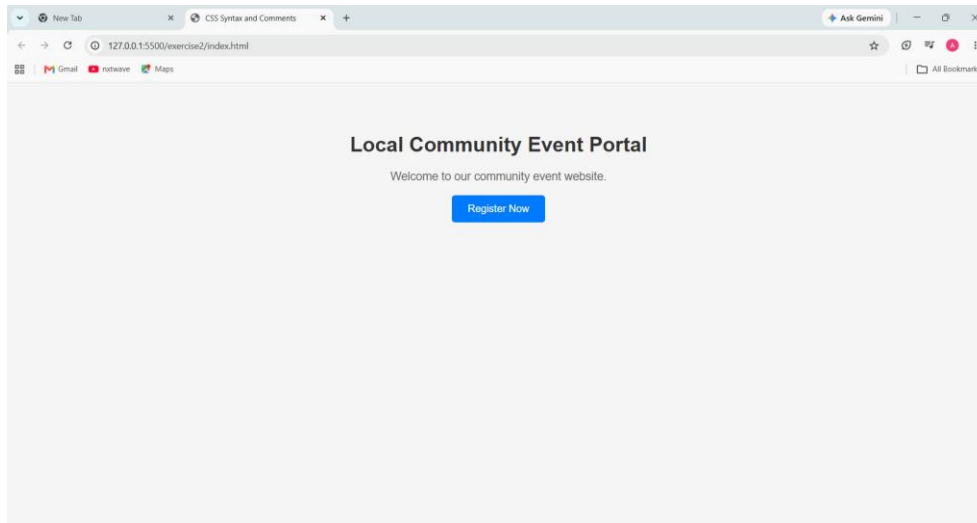
- Create a section in styles.css with formatted rules and consistent indentation.
- Add descriptive comments above selectors.
- Example: `/* Style for main CTA button */ .cta-button { background-color: #007BFF; color: white`



The screenshot shows a code editor with two files open: `index.html` and `styles.css`. The `index.html` file contains the following code:

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>CSS Syntax and Comments</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <h1>Local Community Event Portal</h1>
10
11   <p>
12     Welcome to our community event website.
13   </p>
14
15   <button class="cta-button">
16     Register Now
17   </button>
18
19 </body>
20 </html>
```

Output:



3. Selectors Playground

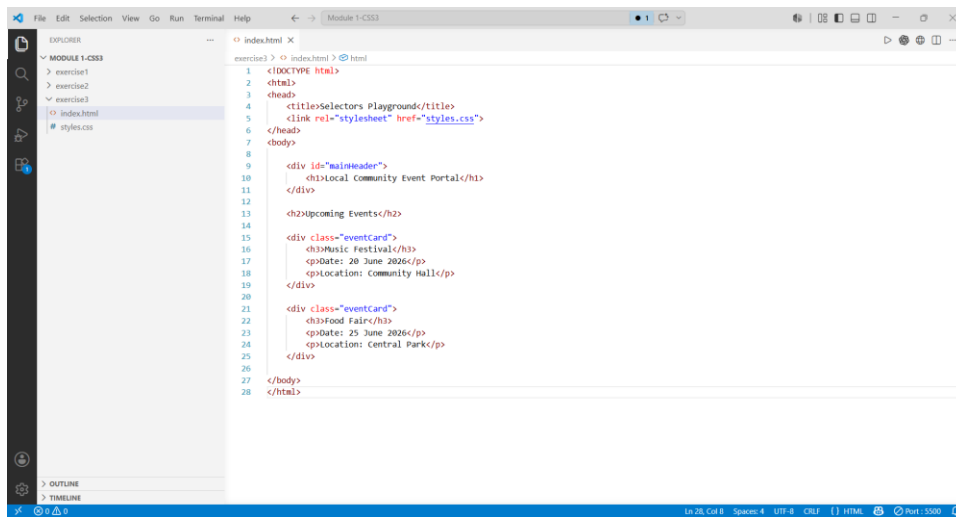
Scenario: You need to style various elements based on IDs, classes, and element types.

Objective: Master different selector types.

Task:

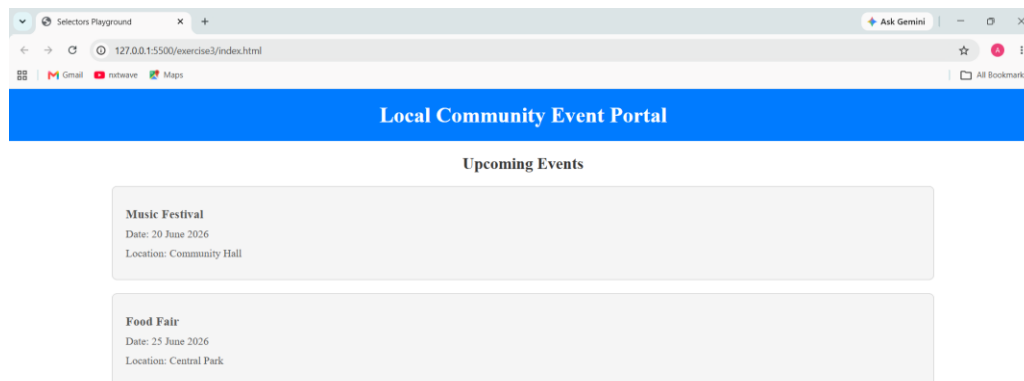
• Use:

- Universal selector * to reset margin/padding
- Element selector to style all
- ID selector #mainHeader for the banner
- Class selector. eventCard for event containers
- Grouping selector for h3, p to style together



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Selectors Playground</title>
5 <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9 <div id="mainHeader">
10 <h1>Local Community Event Portal</h1>
11 </div>
12
13 <h2>Upcoming Events</h2>
14
15 <div class="eventCard">
16 <h3>Music Festival</h3>
17 <p>Date: 20 June 2026</p>
18 <p>Location: Community Hall</p>
19 </div>
20
21 <div class="eventCard">
22 <h3>Food Fair</h3>
23 <p>Date: 25 June 2026</p>
24 <p>Location: Central Park</p>
25 </div>
26
27 </body>
28 </html>
```

Output:



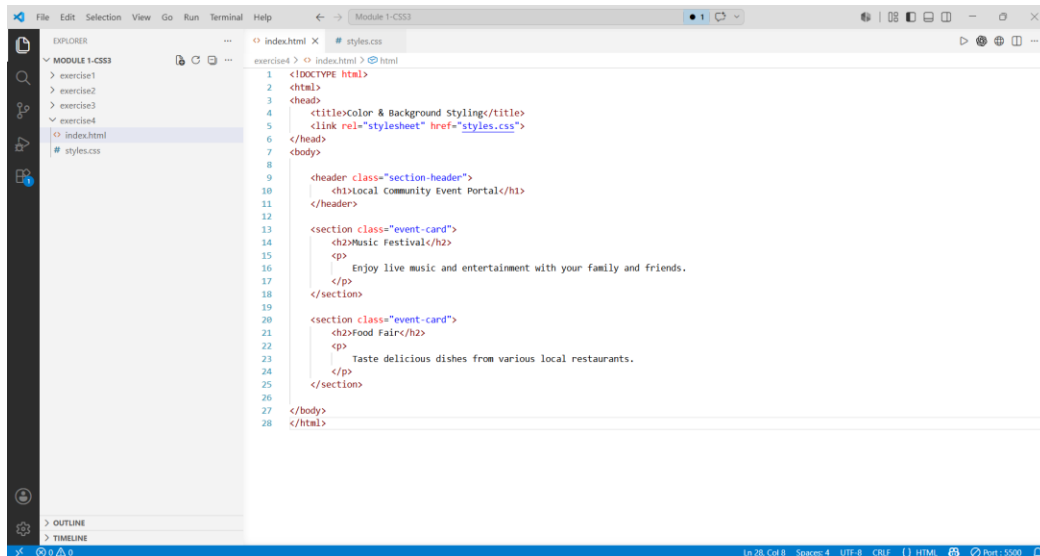
4. Color & Background Styling

Scenario: You're theming the portal based on a city council's branding.

Objective: Apply consistent colors and background visuals

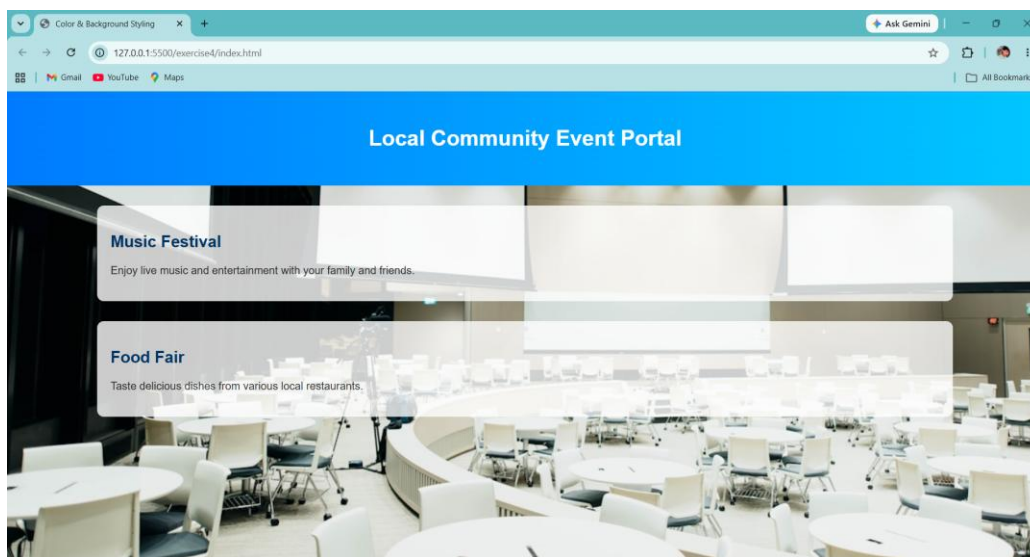
. Task:

- Use HEX and RGBA for setting text and background colors
- Add a background image to the body with fallback color
- Apply gradients to section headers using background: linear-gradient(...)



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Color & Background Styling</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <header class="section-header">
10     <h1>Local Community Event Portal</h1>
11   </header>
12
13   <section class="event-card">
14     <h2>Music Festival</h2>
15     <p>
16       Enjoy live music and entertainment with your family and friends.
17     </p>
18   </section>
19
20   <section class="event-card">
21     <h2>Food Fair</h2>
22     <p>
23       Taste delicious dishes from various local restaurants.
24     </p>
25   </section>
26
27 </body>
28 </html>
```

Output:



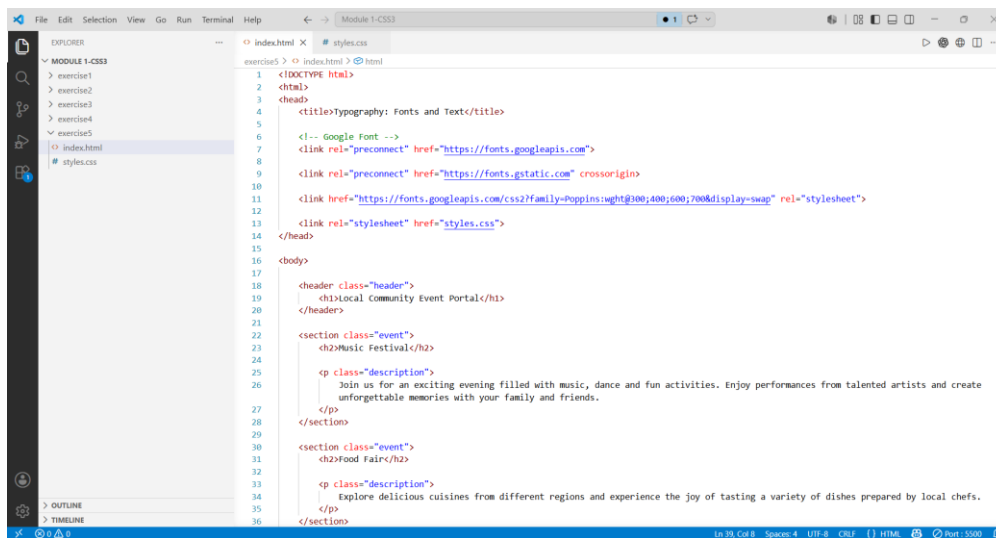
5. Typography: Fonts and Text

Scenario: The marketing team wants more appealing fonts and better readability.

Objective: Enhance textual appearance using CSS properties.

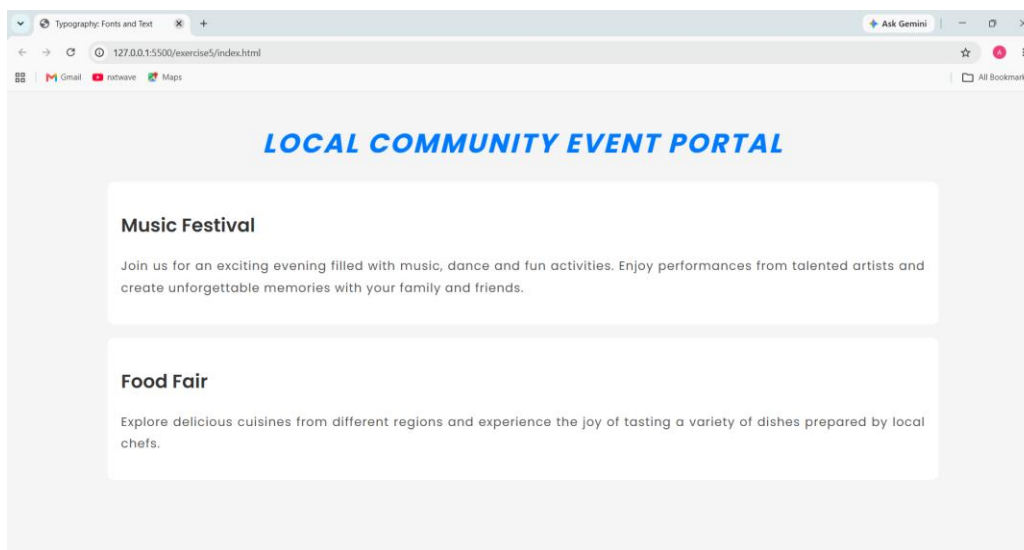
Task:

- Use `@import` or to include a Google Font
- Set font-family, font-size, font-style, font-weight in different sections
- Use text-align, text-transform, letter-spacing, line-height on descriptions



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>Typography: Fonts and Text</title>
5
6 <!-- Google Font -->
7 <link rel="preconnect" href="https://fonts.googleapis.com">
8
9 <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
10
11 <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600;700&display=swap" rel="stylesheet">
12
13 <link rel="stylesheet" href="styles.css">
14 </head>
15
16 <body>
17
18 <header class="header">
19 <h1>Local Community Event Portal</h1>
20 </header>
21
22 <section class="event">
23 <h2>Music Festival</h2>
24
25 <p class="description">
26   Join us for an exciting evening filled with music, dance and fun activities. Enjoy performances from talented artists and create
27   unforgettable memories with your family and friends.
28 </p>
29 </section>
30
31 <section class="event">
32 <h2>Food Faire</h2>
33
34 <p class="description">
35   Explore delicious cuisines from different regions and experience the joy of tasting a variety of dishes prepared by local chefs.
36 </p>
37 </section>
```

Output:



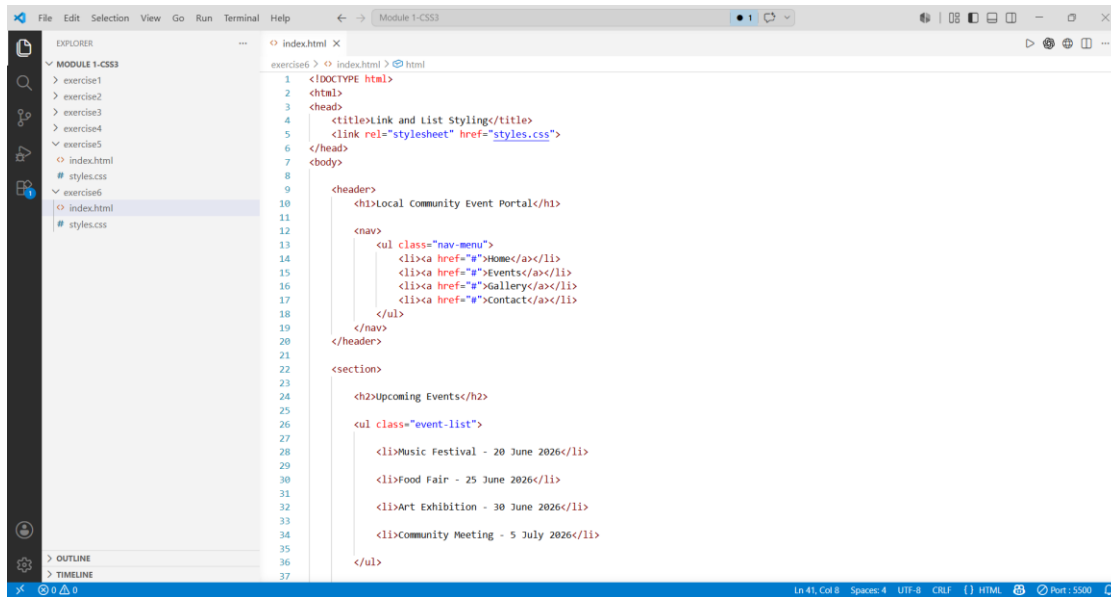
6. Link and List Styling

Scenario: The default blue links and bullet lists don't match the design.

Objective: Customize links and lists.

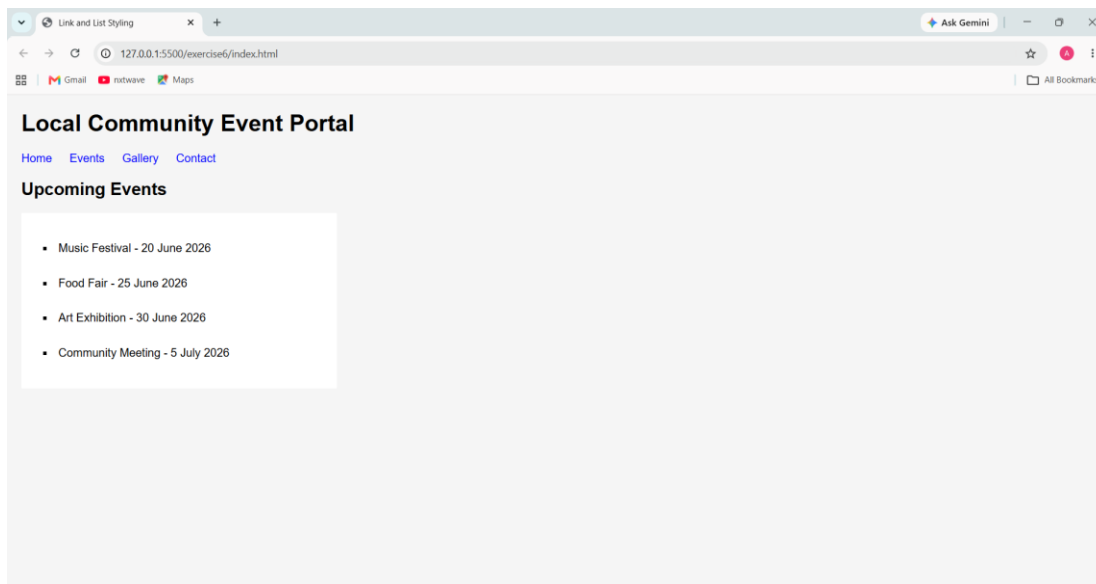
Task:

- Style links with `:link`, `:hover`, `:active`, and `:visited` pseudo-classes
- Use `list-style-type`, `list-style-position`, and remove bullets from nav menus
- Add padding and margin to list items for spacing



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Link and List Styling</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <header>
10    <h1>Local Community Event Portal</h1>
11
12    <nav>
13      <ul class="nav-menu">
14        <li><a href="#">Home</a></li>
15        <li><a href="#">Events</a></li>
16        <li><a href="#">Gallery</a></li>
17        <li><a href="#">Contact</a></li>
18      </ul>
19    </nav>
20  </header>
21
22  <section>
23
24    <h2>Upcoming Events</h2>
25
26    <ul class="event-list">
27      <li>Music Festival - 20 June 2026</li>
28      <li>Food Fair - 25 June 2026</li>
29      <li>Art Exhibition - 30 June 2026</li>
30      <li>Community Meeting - 5 July 2026</li>
31    </ul>
32  </section>
33
34
35
36
37
```

Output:



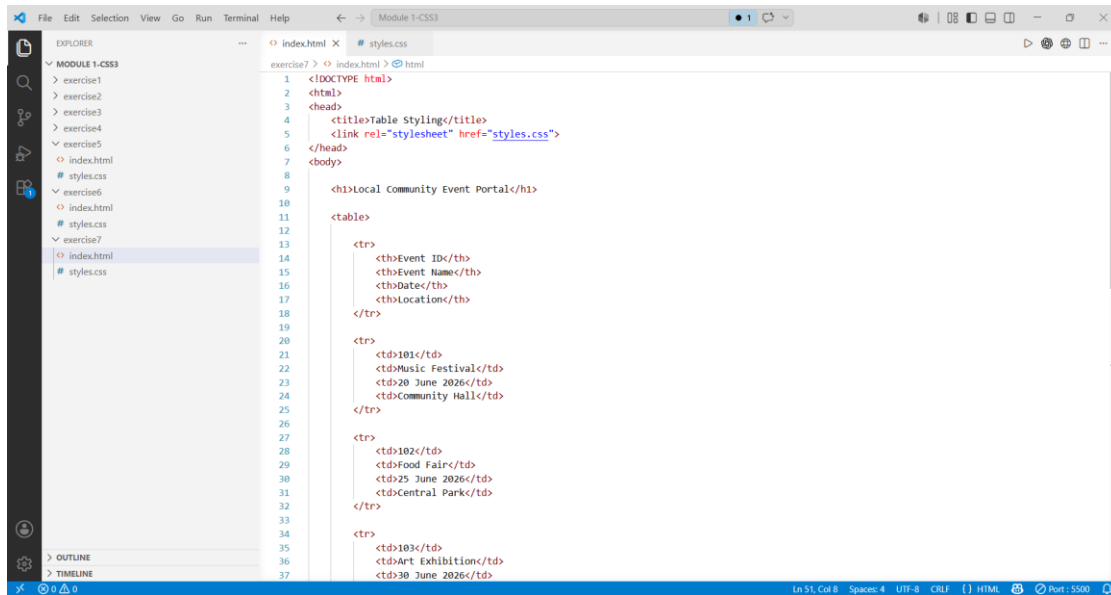
7. Table Styling

Scenario: The events admin table needs a cleaner look.

Objective: Format tables using CSS.

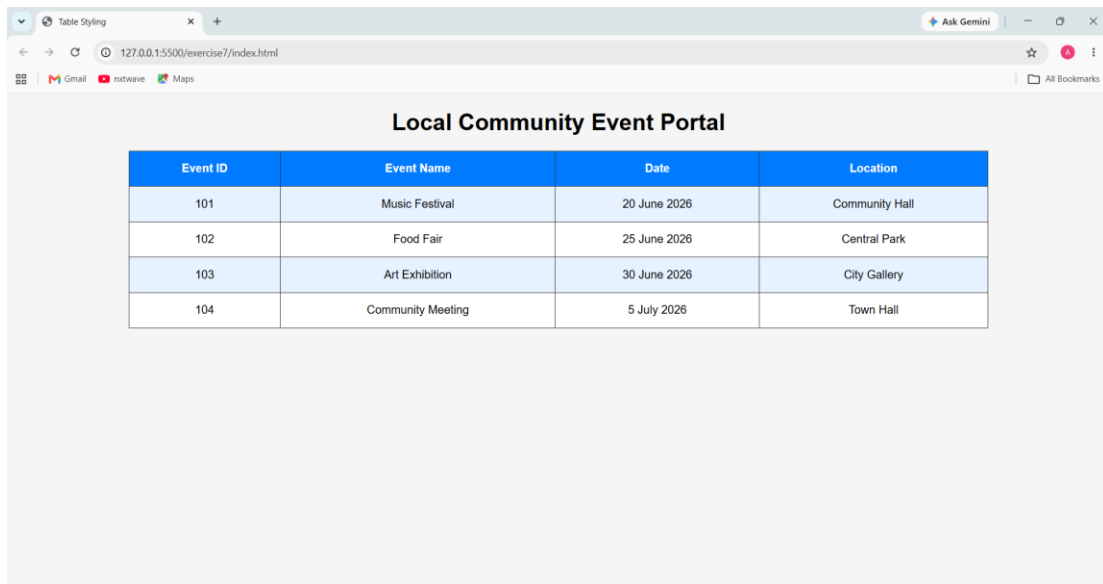
Task:

- Style table, th, and td with borders, padding, and background color
- Add zebra striping to rows using nth-child(even)
- Use border-collapse: collapse and text-align: center



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Table Styling</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <h1>Local Community Event Portal</h1>
10
11   <table>
12
13     <tr>
14       <th>Event ID</th>
15       <th>Event Name</th>
16       <th>Date</th>
17       <th>Location</th>
18     </tr>
19
20     <tr>
21       <td>101</td>
22       <td>Music Festival</td>
23       <td>20 June 2026</td>
24       <td>Community Hall</td>
25     </tr>
26
27     <tr>
28       <td>102</td>
29       <td>Food Fair</td>
30       <td>25 June 2026</td>
31       <td>Central Park</td>
32     </tr>
33
34     <tr>
35       <td>103</td>
36       <td>Art Exhibition</td>
37       <td>30 June 2026</td>
```

Output:



Event ID	Event Name	Date	Location
101	Music Festival	20 June 2026	Community Hall
102	Food Fair	25 June 2026	Central Park
103	Art Exhibition	30 June 2026	City Gallery
104	Community Meeting	5 July 2026	Town Hall

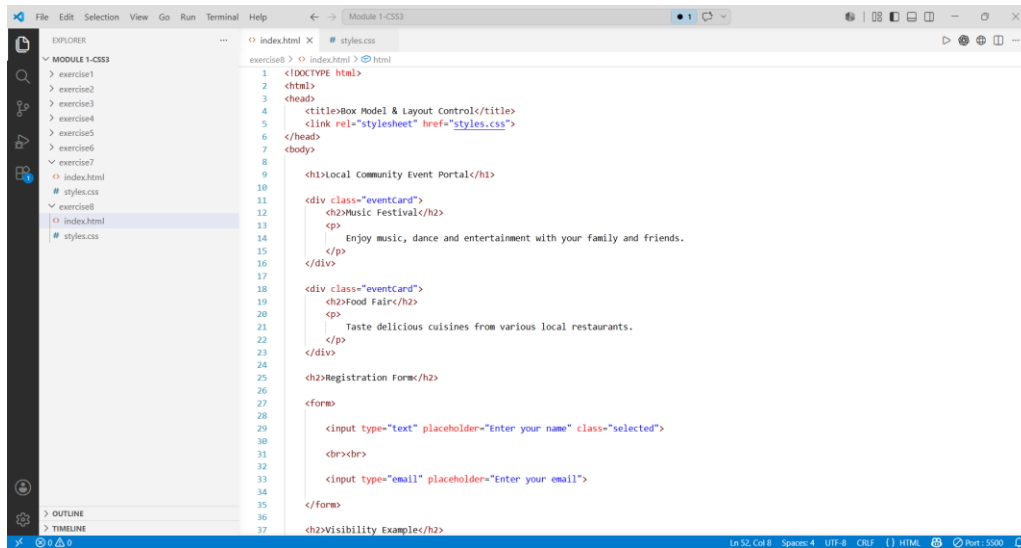
8. Box Model & Layout Control

Scenario: Sections are cramped and need spacing.

Objective: Control element spacing with margin, padding, border, and outline.

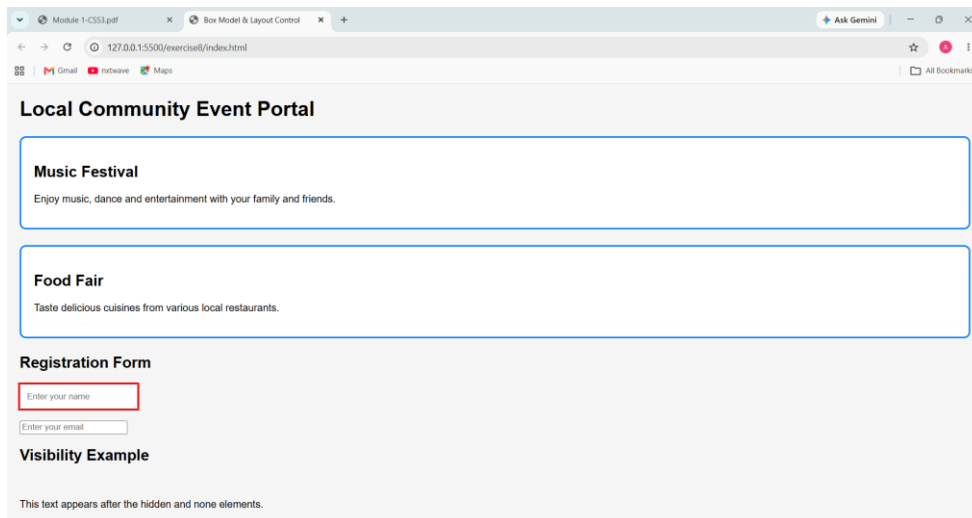
Task:

- Use developer tools to inspect and tweak box model properties
- Add border, padding, and margin to .eventCard
- Add outline to highlight selected fields in a form
- Compare visibility: hidden vs. display: none



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Box Model & Layout Control</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <h1>Local Community Event Portal</h1>
10
11   <div class="eventCard">
12     <h2>Music Festival</h2>
13     <p>
14       Enjoy music, dance and entertainment with your family and friends.
15     </p>
16   </div>
17
18   <div class="eventCard">
19     <h2>Food Fair</h2>
20     <p>
21       Taste delicious cuisines from various local restaurants.
22     </p>
23   </div>
24
25   <h2>Registration Form</h2>
26
27   <form>
28
29     <input type="text" placeholder="Enter your name" class="selected">
30
31     <br><br>
32
33     <input type="email" placeholder="Enter your email">
34
35   </form>
36
37   <h2>Visibility Example</h2>
```

Output:



9. Multiple Columns in Text

Scenario: The community bulletin needs to be displayed like a newspaper.

Objective: Use CSS3 multi-column layout.

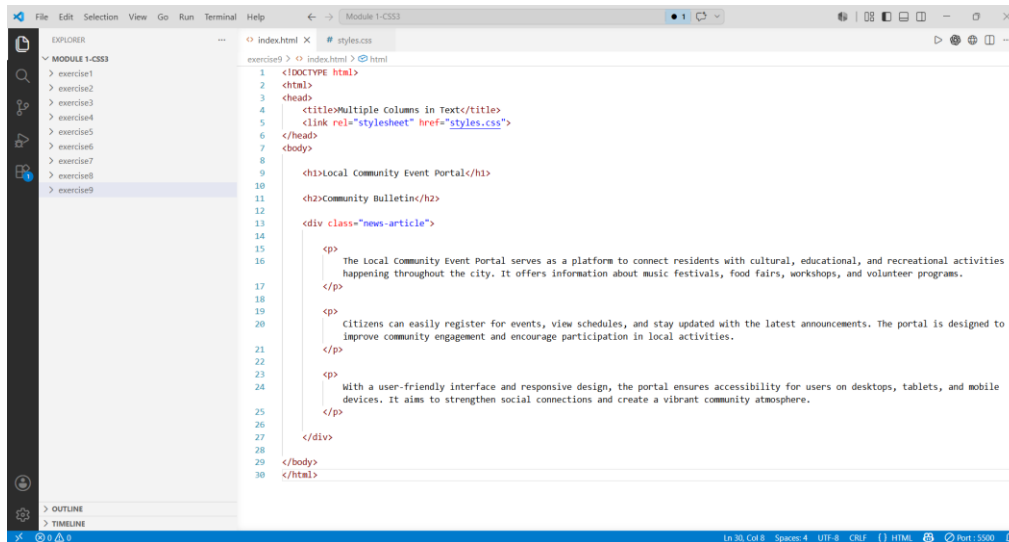
Task:

- Create a news article section and apply

column-count: 2;

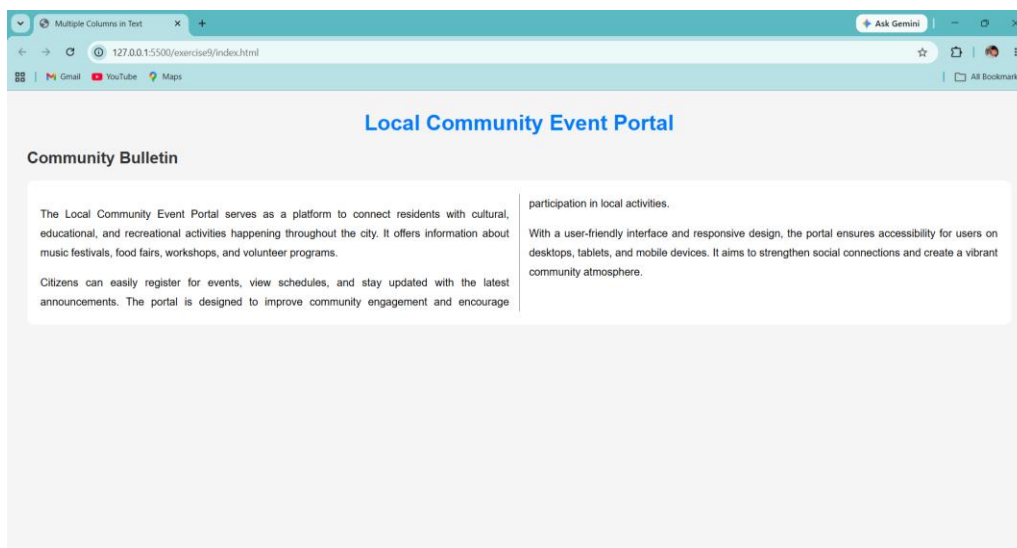
column-gap: 30px;

column-rule: 1px solid gray;



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Multiple Columns in Text</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <h1>Local Community Event Portal</h1>
10
11   <h2>Community Bulletin</h2>
12
13   <div class="news-article">
14
15     <p>
16       The local Community Event Portal serves as a platform to connect residents with cultural, educational, and recreational activities
17       happening throughout the city. It offers information about music festivals, food fairs, workshops, and volunteer programs.
18     </p>
19
20     <p>
21       Citizens can easily register for events, view schedules, and stay updated with the latest announcements. The portal is designed to
22       improve community engagement and encourage participation in local activities.
23     </p>
24
25     <p>
26       With a user-friendly interface and responsive design, the portal ensures accessibility for users on desktops, tablets, and mobile
27       devices. It aims to strengthen social connections and create a vibrant community atmosphere.
28     </p>
29   </div>
30 </body>
31 </html>
```

Output:



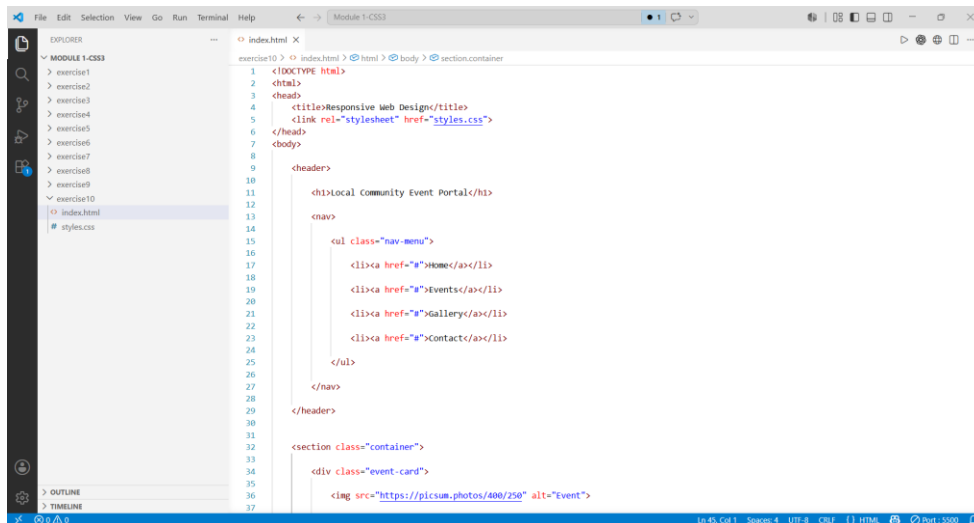
10. Responsive Web Design with Media Queries

Scenario: Users will access the portal on phones, tablets, and desktops.

Objective: Apply media queries for responsiveness.

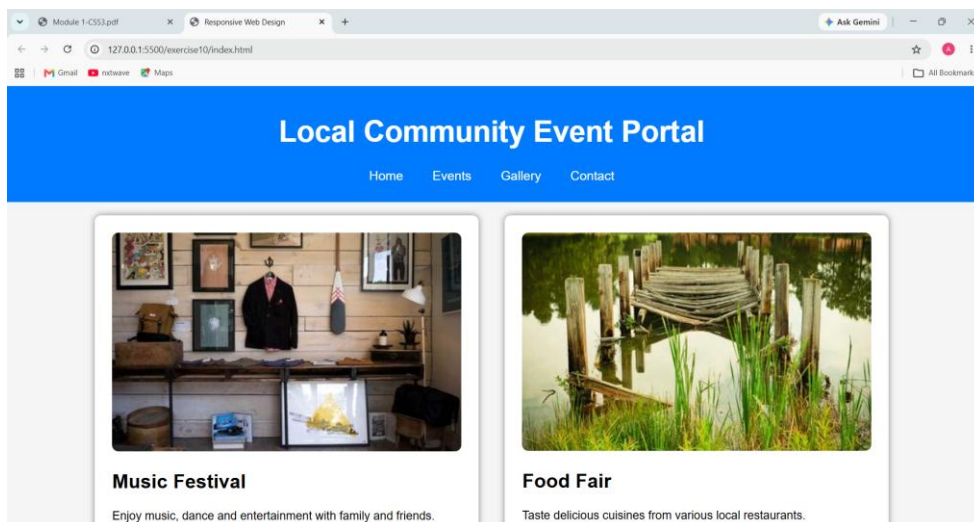
Task:

- Add a media query for screens smaller than 768px
- Stack navigation links vertically instead of horizontally
- Reduce image sizes and font sizes
- Use %, vw, vh for flexible layouts
- Bonus: Try Flexbox or Grid for responsive layouts



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Responsive Web Design</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7 <body>
8
9   <header>
10
11     <h1>Local Community Event Portal</h1>
12
13     <nav>
14
15       <ul class="nav-menu">
16
17         <li><a href="#">Home</a></li>
18
19         <li><a href="#">Events</a></li>
20
21         <li><a href="#">Gallery</a></li>
22
23         <li><a href="#">Contact</a></li>
24
25       </ul>
26
27     </nav>
28
29   </header>
30
31   <section class="container">
32
33     <div class="event-card">
34
35       
36
37     </div>
38
39   </section>
40 </body>
41 </html>
```

Output:



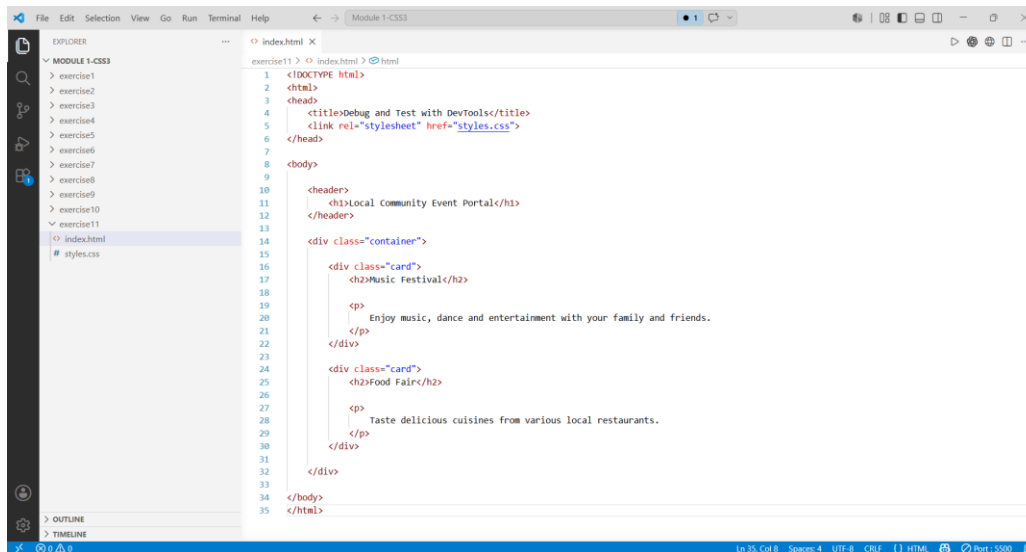
11. Debug and Test with Dev Tools and VS Code

Scenario: Some users report layout breaking on smaller screens.

Objective: Use DevTools to test styles and debug.

Task:

- Use Chrome's device toolbar to simulate different screen sizes
- Inspect applied styles and test different values live
- Use the Network tab to check if the external CSS is loaded



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>Debug and Test with DevTools</title>
5   <link rel="stylesheet" href="styles.css">
6 </head>
7
8 <body>
9
10  <header>
11    <h1>Local Community Event Portal</h1>
12  </header>
13
14  <div class="container">
15
16    <div class="card">
17      <h2>Music Festival</h2>
18
19      <p>
20        Enjoy music, dance and entertainment with your family and friends.
21      </p>
22    </div>
23
24    <div class="card">
25      <h2>Food Fair</h2>
26
27      <p>
28        Taste delicious cuisines from various local restaurants.
29      </p>
30    </div>
31  </div>
32
33 </body>
34 </html>
35
```

Output:

